DEPI INDUSTRY DELIVERABLE — MODULE 3

Requirements Gathering

Project: Automated UI Testing for an E-Commerce Website

1. Stakeholder Analysis

STAKEHOLDER	ROLE	KEY NEEDS & EXPECTATIONS
End Users / Customers	Primary Platform Users	Smooth, defect-free checkout and browsing experience across standard browsers.
QA Automation Team	Framework Developers	Maintainable test scripts, clean POM structure, and unambiguous pass/fail reporting.
Development Team	Software Engineers	Rapid regression feedback to identify UI breakages introduced by new builds.
Academic Evaluator	Project Supervisor	Complete documentation according to DEPI guidelines and full core flow test coverage.

2. Environment & Technical Requirements

- **Programming Language:** Python or Java runtime environment installed and configured.
- **Testing Framework:** Integration of Selenium WebDriver library alongside a test runner (PyTest or TestNG).
- **Browser Drivers:** Web browser drivers (e.g., ChromeDriver) configured in system PATH.
- **IDE:** Suitable IDE (VS Code, PyCharm, or IntelliJ IDEA) for script development and debugging.

3. Functional Testing Requirements (Target Journeys)

- **Authentication Flow:** Verification of login mechanism using valid/invalid credentials and successful page redirection.
- **Product Search & Selection:** Locating target items and adding them to the shopping cart.
- **Cart Management:** Verifying dynamic cart updates (item counter badge assertions) when items are added or removed.
- **Checkout Process:** Navigating checkout, entering synthetic billing/shipping info, and verifying order confirmation.

4. User Stories & Use Cases

> **US-1 (Authentication):** As an e-commerce customer, I want to log in with valid credentials so that I can access my account.

> **US-2 (Cart Management):** As a shopper, I want to select products and add them to my cart so that I can review my order.

> **US-3 (Order Checkout):** As a buyer, I want to enter my shipping info and complete checkout so that my order is confirmed.

5. Non-Functional Requirements

- **Performance & Efficiency:** Fast execution using dynamic explicit waits instead of hardcoded sleep delays.
- **Reliability & Stability:** Zero false-negative failures caused by dynamic element loading or timing mismatches.
- **Maintainability:** Adherence to Page Object Model (POM) design pattern for centralized locator updates.
- **Data Security:** Exclusive use of synthetic/dummy test data to avoid exposing real user information.

6. Test Data Requirements

- **User Credentials:** Standard valid test accounts (e.g., `standard_user`) and invalid accounts for negative testing.
- **Dummy Customer Information:** Synthetic first name, last name, and postal codes for checkout verification.